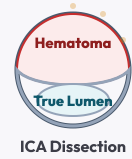
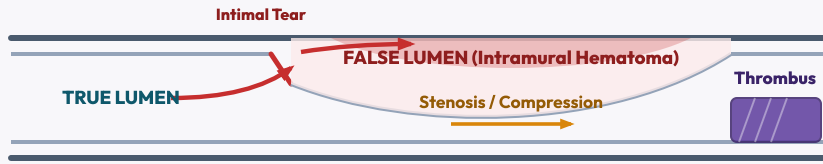


Cervical Artery Dissection

Clinical presentation, diagnostic workup, medical management (extracranial vs. intracranial), and landmark trial evidence.



1. Clinical Presentation & Pathophysiology

Local Pain Profiles (Ipsilateral)

- **Carotid (ICA) Dissection:** Characteristically frontotemporal, retro-orbital, or facial pain (angle of jaw).
- **Vertebral (VA) Dissection:** Severe occipital headache or posterior upper neck pain.
- **Clinical Pearl:** Sudden local headache/neck pain precedes TIA or ischemic stroke by hours to days (median 4 days).

Sympathetic Horner Syndrome

- **Presentation Rate:** ~28–58% in ICA dissection.
- **Sympathetic Sparing:** Presents as ptosis & miosis **WITHOUT** facial anhidrosis.
- **Mechanism:** Sudomotor (sweat) fibers branch at bifurcation and ascend with **external carotid artery (ECA)**, leaving them spared in ICA dissection.

Neurological Deficits

- **Lower Cranial Nerves:** CN IX, X, XI, XII palsies (~8–16%) due to local compression or ischemic *vasa nervorum* injury.
- **Posterior Circulation (VA):** Wallenberg syndrome (lateral medullary), PICA/AICA territory strokes, or cerebellar ataxia.

2. Diagnostic Workup

- **CTA Head & Neck:** Rapid first-line tool. Key signs: tapered stenosis ("string sign"), dissecting pseudoaneurysm, intimal flap, or occlusion.
- **MRI Neck (T1 Fat-Sat):** Pathognomonic hyperintense crescentic wall hematoma (crescent sign; highly sensitive subacutely).
- **DSA:** Gold standard for equivocal cases, stenting assessment, or dynamic flow evaluation.
- **Screening:** Screen for FMD or collagen vascular disorders (Marfan, vascular Ehlers-Danlos) if recurrent or spontaneous without trauma.

3. Medical Management: Extracranial vs. Intracranial Dissection

Extracranial Dissection

- **Antithrombotics:** Indicated for ≥ 3 months (Class I, LOE C-LD).
- **Antiplatelet vs. Anticoagulation:** Clear clinical equipoise. Monotherapy (Aspirin 81–325mg) or DAPT (Aspirin+Clopidogrel) vs. VKA (INR 2.0–3.0) or DOAC.
- **STOP-CAD Transition:** In occlusive dissection, consider early anticoagulation for the high-stroke-risk **first 30 days**, then switch to antiplatelet therapy to limit cumulative bleeding.
- **IV Thrombolysis (IVT):** Safe and indicated for acute stroke within 4.5 hours (Class I).

Intracranial & Pseudoaneurysms

- **Intracranial SAH Risk:** Intracranial vessels lack an external elastic lamina and have thin adventitia. Dissection carries a high rupture risk.
- **Anticoagulation Caution:** Anticoagulation is contraindicated if SAH is present. Perform CT Head/LP first. Prefer single antiplatelet if no SAH.
- **Pseudoaneurysm:** Asymptomatic extracranial pseudoaneurysms are managed conservatively with repeat imaging.
- **Stenting:** Reserve for refractory TIA/stroke or symptomatic enlarging pseudoaneurysms.

4. Landmark Trial & Cohort Evidence

Study / Year	Population & Design	Interventions Compared	Key Outcomes & Clinical Nuance
CADISS 2015	N = 250 patients with acute extracranial dissection (96% carotid, 4% vertebral). RCT.	Antiplatelet (n=126) vs. Anticoagulant (n=124) for 3 months.	Primary Composite (Stroke/Death at 3 Months): Antiplatelet 2.0% (2/126) vs. Anticoagulant 1.0% (1/124) (\$p=0.63\$). No significant difference; established clinical equipoise.
TREAT-CAD 2021	N = 194 randomized (PP analyzed: 173). Acute extracranial dissection. RCT.	Aspirin 300mg daily (n=91) vs. VKA (INR 2.0–3.0, n=82) for 3 months.	Composite Endpoint (Stroke, bleed, death, or MRI at 14d): Aspirin 23% vs. VKA 15% (Diff 8%, upper 95% CI 21% exceeded 12% margin, non-inferiority NOT met). Ischemic stroke alone: Aspirin 8.0% (7/91) vs. VKA 0% (0/82).
Kaufmann IPD 2024	N = 444. Individual patient data meta-analysis combining CADISS & TREAT-CAD.	Antiplatelet vs. Anticoagulant therapy for 3 months.	Composite 90-day Stroke/Death/Major Bleed: No difference (4% vs. 1%). Ischemic Stroke alone: Anticoagulation showed significant reduction (0.5% vs. 4.0%; OR 0.14, 95% CI 0.02–0.61, p=0.01).
STOP-CAD 2024	N = 3,636. Large multicenter retrospective cohort study of cervical dissection.	Antiplatelets vs. Anticoagulation in real-world clinical practice.	Ischemic Stroke & Bleed Risks: Anticoagulation associated with non-significant lower stroke rate but higher bleeding rate. Occlusive subgroup derived significant ischemic benefit. Transition strategy: Use anticoagulation for high-stroke 30-day window, then switch to antiplatelets.

CADISS: *Lancet Neurol.* 2015;14(4):361-7. PMID: 25684164 | **TREAT-CAD:** *Lancet Neurol.* 2021;20(5):341-350. PMID: 33765420
Kaufmann IPD: *JAMA Neurol.* 2024;81(6):630-637. PMID: 38739383 | **STOP-CAD:** *Stroke.* 2024;55(4):908-918. PMID: 38334460 | **AHA/ASA:** *Stroke.* 2021;52:e364-e467. PMID: 34024117